Reinforcement Learning for Noise-Aware Quantum Circuit Compilation

Team Members:
Emails:

1 Objective

Quantum algorithms are typically designed and compiled under the assumption of ideal, noise-free execution. However, in current Noisy Intermediate-Scale Quantum (NISQ) devices, hardware noise, limited qubit connectivity, and gate-dependent error rates significantly degrade the performance of executed circuits. As a result, we often see that circuits which are optimal in theory perform poorly when deployed on real or simulated quantum hardware.

The objective of our project is to investigate whether deep RL algorithms discussed in class can be used and extended to construct quantum circuits that are explicitly optimized for noisy execution rather than idealized correctness. We plan to work with proximal policy optimization, although this is subject to change based on observed performance. Specifically, we aim to formulate this quantum circuit compilation as a sequential decision-making problem in which an agent incrementally constructs a circuit and receives reward based on the fidelity (i.e., how close actual output state is to theoretical expected output state) of the executed circuit under a realistic noise model. We hypothesize that such an approach can yield circuits that are much more robust to noise than standard compilation techniques.

This problem is important because it addresses a key bottleneck in near-term quantum computing: the mismatch between theoretical circuit design and physical hardware execution. Improving noise robustness at the compilation level could meaningfully extend the practical capabilities of NISQ devices without requiring improvements in hardware quality.

2 Related Work

This work takes inspiration from AlphaChip (1) in classical computing, which pioneered using RL for macro placement in classical chip design by treating chip placement as a sequential decision problem to improve the power, performance, and area of the completed chip. AlphaChip trained a graph neural network (GNN) to optimize wirelength, congestion, and area. The results of the paper have been subject to debate in the academic community, but nonetheless showed that RL agents can discover unqiue design strategies for chip-design. Our work aims to adapt this idea for quantum compilation, optimizing for an objective function which minimizes quantum noise.

Other studies have more broadly attempted to solve for quantum error corection such as Nautrup et al. (2), which showed showed that RL can be used to optimize existing surface code quantum memories. Ollé et al. (3) similarly worked to find QEC codes and their encoding circuits for specified gate sets and error models. Both papers focus on discovering protective code structures, but current work has not yet solved for arbitrary target computations (unitaries) on near-term Noisy Intermediate-Scale Quantum hardware.

Pozzi et al. (4) and other papers showed that Qubit routing (inserting SWAP gates, which swap the quantum states of qubits, to minimize added depth while satisfying circuit constraints) could be interpreted as an RL problem, and further, the RL agent can outperform routing procedures in many scenarios. However, these studies have worked under the assumption of a fixed logical circuit, focussing rather on mapping to hardware. Instead, our work constructs the gate sequence itself. This allows the agent to discover circuits whose structure is intrinsically robust to noise.

3 Technical Outline

We formulate quantum circuit compilation as a Markov decision process, where the state consists of a partially constructed circuit and the action space corresponds to the adding another quantum gate to the circuit from a predefined, finite set of possible gates. One episode terminates when the agent outputs a STOP action or reaches a maximum circuit depth.

The reward is computed based on the fidelity between the target unitary (i.e., target operation) and the noisy execution of the constructed circuit. We simulate noise using standard models such as depolarizing and amplitude damping noise within a quantum simulation framework (most likely Qiskit). We may also potentially include a complexity penalty proportional to circuit depth to encourage efficient solutions.

We will train a policy using Proximal Policy Optimization (PPO), allowing the agent to learn a stochastic policy over circuit construction steps. Our key technical contribution is reformulating circuit compilation not as some optimization task over a fixed set of possible circuit representations but instead a sequential decision problem (which is where RL comes in).

We will then evaluate our agent on small-scale quantum circuits (2–3 qubits), comparing against standard compilation baselines and random circuit generation. Performance is measured in terms of fidelity, circuit depth, and robustness across varying noise strengths and unseen noise models. We will then look deeper into failure modes for our agent, including potential overfitting to specific noise distributions and structural biases in learned circuits.

4 Team Contributions

- **Team Member 1**: Responsible for implementing the quantum circuit simulation environment, including construction of target circuits, noise models, and fidelity evaluation using a quantum computing framework (most likely IBM Qiskit). Also contributes to baseline implementations and evaluation infrastructure. Will work alongside team member 2 to implement the PPO agent.
- **Team Member 2**: Primary team member responsible for designing and implementing the reinforcement learning agent using the PPO algorithm, including state/action representation, reward design, and training pipeline. Also leads data analysis, including interpretation of any new information revealed about quantum execution from the agent's learned approach.

References

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